

- * Description : Converts the temperature in DegF
- * protocol to the format
- * expected by the cluster attribute
- * Measured Value in the
- * Temperature Measurement cluster.
- * The half deg F format is 8-bit
- * unsigned, and represents
- * 2x temperature value in
- * Fahrenheit (to get 0.5 resolution).
- * The format expected by cluster
- * is 16-bit signed in Celsius and
- * multiplied by 100 to get
- * 0.01 resolution.
- * e.g. 0x9A (77 deg F) ---> 2500 (25.00 deg C)
- * Input Para : temperature in DegF format
- * Output Para: temperature in Matter format
- */

```
int16s translateDegFTemp(int8u temperature)
{
    int32s x = temperature;

    // rearrangement of
    // = 100 * (x/2 - 32) * 5/9
    // *1000 (should be 100), +90, then /10,
    // is for rounding.

    return (int16s) (((x - 64)*5*1000 + 90) / (10*2*9));
}
```

7.19.2.10. Power Type

This type, derived from int64, represents power measured in milliwatts.

7.19.2.11. Amperage Type

This type, derived from int64, represents amperage measured in milliamps.

7.19.2.12. Voltage Type

This type, derived from int64, represents voltage measured in millivolts.

7.19.2.13. Energy Type

This type, derived from int64, represents energy measured in milliwatt-hours.